Probabilistic Combinatorics (Math 5390) Fall 2026
Shagnik Das

Midterm Practice — Solutions

Throughout, Pr and $\mathbb{E}$ refer to the random experiment described in each solution, and $|\cdot|$ denotes cardinality.

Problem 1.2 *Suppose $n \geq 4$, and let H be an n-uniform hypergraph with at most $4^{n-1}/3^n$ edges. Prove that there is a coloring of the vertices of H by four colors so that in every edge all colors are represented.*

Solution. If H has no edges there is nothing to prove, so assume $1 \leq |E(H)| \leq 4^{n-1}/3^n$. Color every vertex independently and uniformly at random with one of the colors $1, 2, 3, 4$. Fix an edge e, and for $c \in \{1, 2, 3, 4\}$ let M_c be the event that color c does not appear on e. Since $|e| = n$ and the colors are independent, for any set C of colors with $|C| = j$,

$$\Pr\Big(\bigcap_{c \in C} M_c\Big) = \Big(\frac{4-j}{4}\Big)^n.$$

Call e *bad* if some color is missing on e. By inclusion–exclusion,

$$\Pr(e \text{ bad}) = \Pr\Big(\bigcup_{c=1}^{4} M_c\Big) = 4\Big(\frac{3}{4}\Big)^n - 6\Big(\frac{1}{2}\Big)^n + 4\Big(\frac{1}{4}\Big)^n < 4\Big(\frac{3}{4}\Big)^n,$$

where the strict inequality holds because $6 \cdot 2^{-n} > 4 \cdot 4^{-n}$. By the union bound over the edges,

$$\Pr(\text{some edge is bad}) \leq \sum_{e \in E(H)} \Pr(e \text{ bad}) < |E(H)| \cdot 4\Big(\frac{3}{4}\Big)^n \leq \frac{4^{n-1}}{3^n} \cdot \frac{4 \cdot 3^n}{4^n} = 1.$$

Hence with positive probability no edge is bad, so some coloring has all four colors on every edge. (The hypothesis $n \geq 4$ is what makes this possible at all: an edge with fewer than 4 vertices cannot see four colors.) □

Problem 1.5 *Let $G = (V, E)$ be a graph on $n \geq 10$ vertices and suppose that if we add to G any edge not in G then the number of copies of a complete graph on 10 vertices in it increases. Show that the number of edges of G is at least $8n - 36$.*

Solution. We use Bollobás' theorem on set pairs.

Lemma 1 (Bollobás)**.** *Let (A_i, B_i), $1 \le i \le h$, be pairs of finite sets with $|A_i| = k$, $|B_i| = \ell$, $A_i \cap B_i = \varnothing$ for all i, and $A_i \cap B_j \neq \varnothing$ for all $i \neq j$. Then $h \le \binom{k+\ell}{k}$.*

Let $e_1, \dots, e_h$ be the non-edges of G, where $h = \binom{n}{2} - |E|$. Adding $e_i = \{x_i, y_i\}$ creates a new copy of K_{10}, which must use the new edge e_i; so there is an 8-set $D_i \subseteq V \setminus e_i$ such that $Q_i = D_i \cup e_i$ spans a K_{10} in $G + e_i$. In other words,

$$\text{every pair inside } Q_i \text{ other than } e_i \text{ is an edge of } G. \tag{1}$$

Set

$$A_i = e_i, \qquad B_i = V \setminus Q_i, \qquad |A_i| = 2, \quad |B_i| = n - 10.$$

Clearly $A_i \cap B_i = \varnothing$ since $e_i \subseteq Q_i$. For $i \neq j$, if $A_i \cap B_j = \varnothing$ then $e_i \subseteq Q_j$ with $e_i \neq e_j$, so e_i is an edge of G by (1), contradicting that e_i is a non-edge. Thus $A_i \cap B_j \neq \varnothing$, and Lemma 1 with $k = 2$, $\ell = n - 10$ gives $h \le \binom{n-8}{2}$. Therefore

$$|E| = \binom{n}{2} - h \ge \binom{n}{2} - \binom{n-8}{2} = \frac{n(n-1) - (n-8)(n-9)}{2} = \frac{16n - 72}{2} = 8n - 36. \qquad \square$$

Problem 1.7 *Let $\{(A_i, B_i), 1 \le i \le h\}$ be a family of pairs of subsets of the set of integers such that $|A_i| = k$ for all i and $|B_i| = \ell$ for all i, $A_i \cap B_i = \varnothing$ for all i and $(A_i \cap B_j) \cup (A_j \cap B_i) \neq \varnothing$ for all $i \neq j$. Prove that $h \le (k+\ell)^{k+\ell}/(k^k \ell^\ell)$.*

Solution. Let $X = \bigcup_i (A_i \cup B_i)$, a finite set, and let $p = \frac{k}{k+\ell}$ (with the convention $0^0 = 1$ if k or ℓ is 0). Color each $x \in X$ independently *left* with probability p and *right* with probability $1 - p$. Let E_i be the event that all of A_i is left and all of B_i is right. As $A_i \cap B_i = \varnothing$, these are $k + \ell$ distinct elements, so

$$\Pr(E_i) = p^k (1-p)^\ell = \frac{k^k \ell^\ell}{(k+\ell)^{k+\ell}}.$$

The E_i are pairwise disjoint: for $i \neq j$ there is some $x \in A_i \cap B_j$ or $x \in A_j \cap B_i$; in the first case E_i forces x left while E_j forces x right, and symmetrically in the second, so $E_i \cap E_j = \varnothing$. Hence

$$h \cdot \frac{k^k \ell^\ell}{(k+\ell)^{k+\ell}} = \sum_{i=1}^{h} \Pr(E_i) = \Pr\Big(\bigcup_i E_i\Big) \le 1,$$

which is the claim. (The choice $p = k/(k+\ell)$ maximizes $p^k(1-p)^\ell$.) $\square$

Problem 1.10 *Prove that there is an absolute constant $c > 0$ with the following property: let A be an $n \times n$ matrix with pairwise distinct entries. Then there is a permutation of the rows of A so that no column in the permuted matrix contains an increasing subsequence of length at least $c\sqrt{n}$.*

Solution. We show $c = 2e$ works. Let $m = \lceil 2e\sqrt{n} \rceil$. If $m > n$ there is nothing to prove, so assume $m \le n$. A column has an increasing subsequence of length $\ge c\sqrt{n}$ iff it has one of length exactly m (lengths are integers, and a subsequence of an increasing sequence is increasing).

Permute the rows by a uniformly random $\sigma \in S_n$, so the new row r is old row $\sigma(r)$. Fix a column j and positions $r_1 < \cdots < r_m$. The tuple $(\sigma(r_1), \ldots, \sigma(r_m))$ is uniform over ordered m-tuples of distinct rows, so conditioned on the *set* $\{\sigma(r_1), \ldots, \sigma(r_m)\}$ its order is uniform among the $m!$ orders. As the entries are distinct, exactly one order makes $A_{\sigma(r_1),j} < \cdots < A_{\sigma(r_m),j}$, so this event has probability $1/m!$. Let N_j be the number of increasing subsequences of length m in column j. Then, using $\binom{n}{m} \le n^m/m!$ and $m! \ge (m/e)^m$ (from $e^m \ge m^m/m!$),

$$\Pr(N_j \ge 1) \le \mathbb{E}[N_j] = \binom{n}{m}\frac{1}{m!} \le \frac{n^m}{(m!)^2} \le \left(\frac{e^2 n}{m^2}\right)^m \le \left(\frac{1}{4}\right)^m,$$

since $m \ge 2e\sqrt{n}$. By the union bound over the n columns, and using $m \ge 2e\sqrt{n}$ and $\ln n \le \sqrt{n}$ for $n \ge 1$,

$$\Pr(\exists j : N_j \ge 1) \le n\, 4^{-m} \le \exp\left(\ln n - 2e \ln 4\sqrt{n}\right) \le \exp\left((1 - 2e\ln 4)\sqrt{n}\right) < 1.$$

So some permutation σ leaves every column without an increasing subsequence of length m, hence of length $\ge c\sqrt{n}$. □

Ramsey numbers.

Definition 1 (Ramsey number)**.** For integers $k, \ell \ge 1$, the *Ramsey number* $R(k, \ell)$ is the least integer N such that every red/blue coloring of the edges of the complete graph K_N contains a red K_k or a blue K_ℓ. Equivalently, every graph on N vertices contains a clique of size k or an independent set of size ℓ. Ramsey's theorem says $R(k, \ell)$ is finite; in particular $R(k, k) > n$ iff there is a red/blue coloring of $E(K_n)$ with no monochromatic K_k.

The bound used below comes from the deletion method: color $E(K_n)$ uniformly at random; each k-set is monochromatic with probability $2 \cdot 2^{-\binom{k}{2}}$, so the expected number of monochromatic K_k is $\binom{n}{k}2^{1-\binom{k}{2}}$. Fix a coloring with at most this many, and delete one vertex from each monochromatic K_k; the remaining $\ge n - \binom{n}{k}2^{1-\binom{k}{2}}$ vertices span a coloring with no monochromatic K_k, so by Definition 1

$$R(k, k) > n - \binom{n}{k}2^{1-\binom{k}{2}} \qquad \text{for every integer } n. \tag{2}$$

Problem 3.1 *As shown in Section 3.1, the Ramsey number $R(k, k)$ satisfies*

$$R(k, k) > n - \binom{n}{k}2^{1-\binom{k}{2}}$$

for every integer n. Conclude that

$$R(k, k) \ge (1 - o(1))\frac{k}{e}2^{k/2}.$$

Solution. Here $o(1)$ is as $k \to \infty$. Take $n = \lfloor \frac{k}{e} 2^{k/2} \rfloor$, so $\frac{en}{k} \le 2^{k/2}$ and $n \ge \frac{k}{e} 2^{k/2} - 1$. Using $\binom{n}{k} \le n^k/k!$ and $k! \ge (k/e)^k$,

$$\binom{n}{k} 2^{1-\binom{k}{2}} \le \Big(\frac{en}{k}\Big)^k 2^{1-\frac{k(k-1)}{2}} \le 2^{\frac{k^2}{2}} \cdot 2^{1-\frac{k^2}{2}+\frac{k}{2}} = 2 \cdot 2^{k/2}.$$

Plugging into (2),

$$R(k,k) > \frac{k}{e} 2^{k/2} - 1 - 2 \cdot 2^{k/2} = \frac{k}{e} 2^{k/2} \Big(1 - \frac{2e}{k} - \frac{e}{k\, 2^{k/2}}\Big) = (1 - o(1)) \frac{k}{e} 2^{k/2},$$

since $\frac{2e}{k} + \frac{e}{k 2^{k/2}} \to 0$. The point is that the deleted term $2 \cdot 2^{k/2}$ is smaller than $n \approx \frac{k}{e} 2^{k/2}$ by a factor $\Theta(1/k)$. □

Problem 3.3 *Prove that every three-uniform hypergraph with n vertices and $m \ge n/3$ edges contains an independent set (i.e., a set of vertices containing no edges) of size at least*

$$\frac{2n^{3/2}}{3\sqrt{3}\sqrt{m}}.$$

Solution. Let H be the hypergraph and $p = \sqrt{n/(3m)}$; the hypothesis $m \ge n/3$ is exactly $p \le 1$. Choose a random set $S \subseteq V(H)$ by including each vertex independently with probability p. Let $Y = |S|$ and let Z be the number of edges $e \subseteq S$. Since each edge has 3 vertices, by linearity of expectation

$$\mathbb{E}[Y - Z] = pn - p^3 m = p(n - p^2 m) = \sqrt{\frac{n}{3m}} \Big(n - \frac{n}{3}\Big) = \frac{2n^{3/2}}{3\sqrt{3}\sqrt{m}}.$$

Fix an outcome with $Y - Z \ge \mathbb{E}[Y - Z]$, and delete from S one vertex of each edge contained in S. The resulting set I has $|I| \ge Y - Z$, and it is independent: an edge inside I would be an edge inside S, one of whose vertices was deleted. Hence $|I| \ge \frac{2n^{3/2}}{3\sqrt{3}\sqrt{m}}$. (The value of p maximizes $pn - p^3 m$: the derivative $n - 3p^2 m$ vanishes at $p^2 = n/(3m)$.) □